

Cover Letter

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Hiring Manager
Zillow
[Company Address]
[City, State, ZIP Code]

Dear Hiring Manager,

I am thrilled to apply for the Principal Software Development Engineer position on Zillow's AI Platform team, where my passion for building scalable AI infrastructure and my extensive experience in machine learning, cloud architectures, and distributed systems align perfectly with your mission to empower millions to find their ideal home. With a Master's in Computer Science from NYU and over 20 years of experience designing and leading production-grade platforms, I am eager to drive Zillow's next-generation AI infrastructure to support innovative products like home value estimation and personalized recommendations.

In my current role as Director of Cloud Architecture at Biophy, I architected a scalable, event-driven AI platform using Python, FastAPI, and Apache Kafka, integrating Retrieval-Augmented Generation (RAG) workflows with AWS (EKS, S3, Lambda) and Terraform for infrastructure management. I led the development of a modular data ingestion pipeline, optimizing PostgreSQL and Elasticsearch for high-throughput processing, which reduced latency by 50% and scaled to handle millions of data points daily. At McKinsey & Company, I spearheaded the migration of legacy systems to a cloud-native architecture using Databricks and Spark, enabling genAI model experimentation and deployment with robust monitoring via AWS CloudWatch. These experiences directly align with Zillow's need for scalable ML infrastructure, leveraging technologies like Kubernetes, Databricks, and Kafka.

My expertise in building low-latency, high-reliability systems equips me to lead Zillow's AI platform initiatives. At Yotascale, I designed a cross-cloud analytics platform using Kubernetes and Istio, ensuring production-quality deployment and observability for microservices. My work with TensorFlow and LangChain at Biophy, combined with CI/CD pipeline development using Jenkins, demonstrates my ability to streamline ML experimentation and production workflows. I also have a strong track record of collaborating with cross-functional teams, as evidenced by my work at McKinsey, where I partnered with data scientists and product managers to align technical solutions with business goals, a critical skill for Zillow's collaborative environment.

As a technical leader, I am passionate about mentoring and fostering innovation. At Biophy, I guided engineers on Clean Architecture and AWS best practices, while at McKinsey, I led workshops on genAI and RAG, elevating team proficiency. My vision for simplifying ML development through reusable frameworks, such as my open-source JSON resume project, aligns with Zillow's goal of making AI accessible for practitioners. I am inspired by Zillow's commitment to reimagining real estate and am eager to bring my technical expertise, leadership, and user-centric approach to build scalable, compliant AI infrastructure that powers impactful products.

Thank you for considering my application. I would welcome the opportunity to discuss how my skills can advance Zillow's AI platform and mission. Please feel free to contact me at 856-408-4027 or rongyj@hotmail.com to schedule a conversation.

Sincerely,
Yongjun Rong